

GelSight × Claude Code — Field Report 🖐️🤖

Can Claude connect to a GelSight tactile sensor on Linux out of the box — or do we crack it like we cracked Tekscan? · 2026-06-04

The big picture. Our Tekscan ELF proof-of-concept showed Claude can read a real-world force sensor on Linux with **literally nothing** — no API, no SDK, no Linux driver — just photos and a from-scratch reverse-engineering of the USB byte protocol. Asking the same of **GelSight**, the answer splits in two: the **GelSight Mini (\$510)** is the **opposite** of Tekscan — an **open, Linux-native Python SDK, plug-and-play** (a one-session job). The bigger **Mobile / Max / Modulus** units are the **same wall** — Windows-only, paid, license-gated. **And nobody — not one product — ships an MCP server or any Claude/LLM-native integration.**

Part 1 — Does GelSight have what would make our life easy?

Question	Answer
Official API / SDK for all sensors?	Partly. GelSight Mini → gsrobotics (Python, GPL-3.0, free, public , ~189★). Industrial Mobile/Max → paid, license-gated .NET Async API or C++ GelSightSDK.
Official MCP server?	None . Zero GelSight presence in the MCP registry, GitHub, or anywhere indexed.
Official Linux software for all sensors?	Only the Mini. Mini = Linux-native (USB UVC webcam → /dev/videoN, Python 3.12, ROS2). Mobile / Max / Modulus = Windows-only GUI .
AI-agent / Claude-native compatibility?	None, for any product . "AI-enhanced metrology" = computer-vision defect detection, <i>not</i> LLMs/agents/MCP. Only path = DIY wrap the open Mini SDK.

Part 2 — Tekscan (what we did) vs the two GelSight paths

	Tekscan ELF (done)		GelSight Mobile/Max (hard)
Linux	❌ Windows-only	✅ native (Python, v4l2, ROS2)	❌ Windows-only GUI
API / SDK	❌ none — RE'd FTDI bytes	✅ open Python SDK (free)	⚠️ paid, license-gated
Connection	FTDI, unknown protocol (we cracked 1 Mbaud + SetFrameRate → StartRecording)	USB webcam → cv2.VideoCapture, zero RE	locked GUI / paid API
Effort for a Claude read-loop	multi-week RE	a single session	contact-sales + Windows
MCP / Claude-native	❌	❌ (DIY)	❌

🔑 **Why the Mini is easy:** it's an 8 MP camera looking up through a coated gel — it enumerates as a plain USB webcam, so there's **no protocol to reverse-engineer**. `pip install -r requirements.txt → v4l2-ctl --list-devices → read frames`. The SDK adds NN depth, 3D point clouds, and marker/shear tracking (.npy/.csv). Calibrated absolute force isn't out-of-the-box, but depth + marker displacement are.

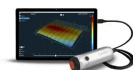
Part 3 — Price & form factor (FlexiForce strip → a bit bigger than a phone)



DIGIT
\$355



Mini ★
\$510



Mobile S2 🎯
Quote



Max
Quote



Modulus
Quote

Product	Price (USD)	Body size	Sensing pad	Size vs anchors	Linux/SDK
DIGIT	\$355	fingertip module	~ fingertip	≈ FlexiForce <i>dot</i>	—
GelSight Mini ★	\$510 (Robotics \$560)	~32×28 mm (coin)	18.6×14.3 mm	smallest , ~2× FF dot	✓ Python/Linux
Mobile Series 2 🎯	Contact sales	grip 158 mm, 400 g	17×14 mm	≈ iPhone (sweet spot)	✗ Win
GelSight Max	Contact sales	67×67×245 mm, 727 g	14.6×8.3 mm	~1.5× iPhone — "a little larger"	✗ Win
GelSight Modulus	Contact sales	handheld, 165 g	swappable optics	phone → larger	✗ Win

Anchors: FlexiForce A201 strip = 191×14×0.2 mm, 9.53 mm dot. iPhone ≈ 147–160 × 71–78 mm. **Note:** nothing GelSight makes is a thin flexible strip — all are 3D camera-in-gel modules. All three handhelds share a tiny ~14–19 mm pad; only the grip/optics scale up.

Recommendation & the opening

- **Buy the GelSight Mini (\$510)** — only publicly-priced full sensor, open Python SDK, plug-and-play on Linux. Skips the entire reverse-engineering odyssey; a Claude-Code read-loop is a single session.
- **iPhone-size sweet spot → Mobile Series 2** — expect "contact sales" + a Windows workflow (the Tekscan-style wall, which our PoC proves we can still crack from nothing).
- 🚀 **The opening: no GelSight product ships an MCP or a Claude-native interface.** Wrapping the open Mini SDK into an MCP server (live tactile depth/force → Claude) is greenfield — exactly the "Claude acts in the real world via sensors" thesis our Tekscan PoC set out to prove.

Sources — GitHub: gelsightinc/g robotics, /apidemo, /sdksdemo. Web: gelsight.com/products, /gelsightmini, /online-store, /product/gelsight-max & -mobile, Mini & Mobile-Series-2 & Max datasheets, "AI-Enhanced Metrology Software" release. Community: joehjhuang/g s_sdk, duyipai/g smini, RVSATHU/gelsight_mini_ros. · Companion files: gelsight_preliminary_product_research.md, gelsight_price_point_analysis.md, gelsight_price_report.pdf.